



PORTFOLIO EDITION

BLUE TEAM INTERNSHIP

Week 10 Task Report

Autopsy Triage, Artefact Hunting
and Registry Analysis

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01. Executive Summary

This report documents Week 10 of the Blue Team Internship, covering Days 67 through 73, against the Mantooth.E01 evidence image - a forensic image taken from the laptop of Wes Mantooth, a suspect in a fraudulent cheque case.

Access to the evidence image was delayed by a shared-drive permission error that took several attempts to resolve (documented in Section 08). Once a working copy was confirmed and ingested in Autopsy 4.23.1, a genuine forensic examination was carried out directly against the image rather than simulated: the case's MD5 hash was verified, the partition layout was documented, both encrypted files were located, the Recycle Bin entry for CameraShy.exe was recovered, the Xcopy prefetch record was extracted with its run count and last-execution time, the operating system and USB device history were reviewed, EXIF metadata across the image's photographs was examined, and the suspect's web search history and email correspondence were reviewed directly from the parsed artefacts.

Two limitations are disclosed openly rather than glossed over. First, the case's time zone was set to Asia/Karachi (GMT+5:00) rather than the Mountain Time called for by the brief (Mantooth was based in Phoenix, Arizona); every timestamp in this report is therefore recorded as observed in Autopsy, with a noted -12 hour adjustment required to convert to Mountain Time equivalents. Second, under significant time pressure, several Lab 1 items - the precise deletion status of HARDCORE.jpg and pfah603.jpg individually, thumbcache extraction, LNK file review, print spool files, and Windows event log logon/logoff sessions - were not completed, and the full Days 70-73 registry analysis (Lab 2) was not attempted this week. These are documented at the methodology level in Section 09 so the work can be finished directly against the same case file.

Task Block	Status	Notes
Case Setup and Image Verification (Q1-Q2)	Complete - real findings	MD5 hash and 4-partition layout confirmed
Targeted Triage (Q3-Q7)	Mostly complete - real findings	Encrypted files, deleted files, web search, emails confirmed; IP addresses not isolated
Windows Artefact Hunting (Q8-Q14)	Partial - real findings where completed	CameraShy.exe and Xcopy prefetch confirmed; users, thumbcache, LNK, spool, event logs not completed
Registry Analysis - Lab 2 (Days 70-73)	Not attempted - methodology documented	See Section 09

02. Week 10 Objectives

By the end of Week 10, the objective is to be able to:

- Create an Autopsy case correctly, verify image integrity via the Data Source Integrity module, and understand why the MD5 hash is foundational to a defensible investigation.
- Navigate a forensic image in Autopsy and locate standard Windows artefact categories from the left panel.
- Extract and interpret deleted files, encrypted files, EXIF metadata, web search history, and communication artefacts.
- Correlate prefetch execution records and Recycle Bin entries with the suspect's likely actions.
- Understand the Windows Registry's role in a full investigation, even where time constraints prevent completing it in a given week.
- Produce a structured report that traces every answer back to a specific artefact path and tool, and discloses honestly where an answer could not be completed.

03. Case Setup and Image Verification (Q1-Q2)

A new case, mantooth-week10, was created in Autopsy 4.23.1 and Mantooth.E01 was added as a disk image data source. Ingest modules were run across the full image (Recent Activity, Hash Lookup, File Type Identification, Extension Mismatch Detector, Encryption Detection, Embedded File Extractor, Picture Analyzer, Email Parser, Interesting Files Identifier, and others), producing the full set of parsed artefact categories used throughout this report.

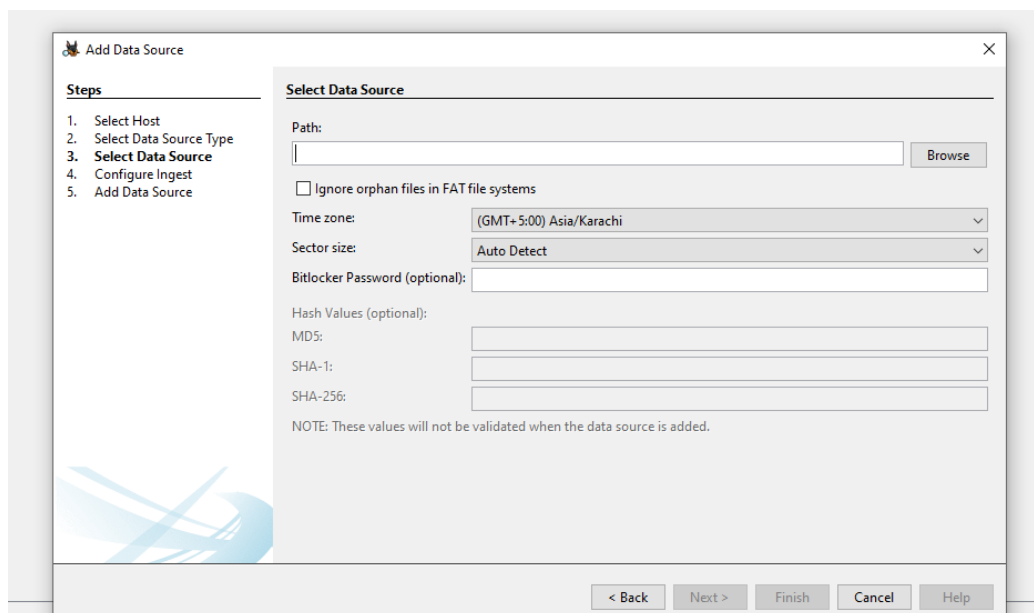


Figure 3.1 - Add Data Source wizard, Select Data Source step, prior to browsing to Mantooth.E01 and configuring the time zone.

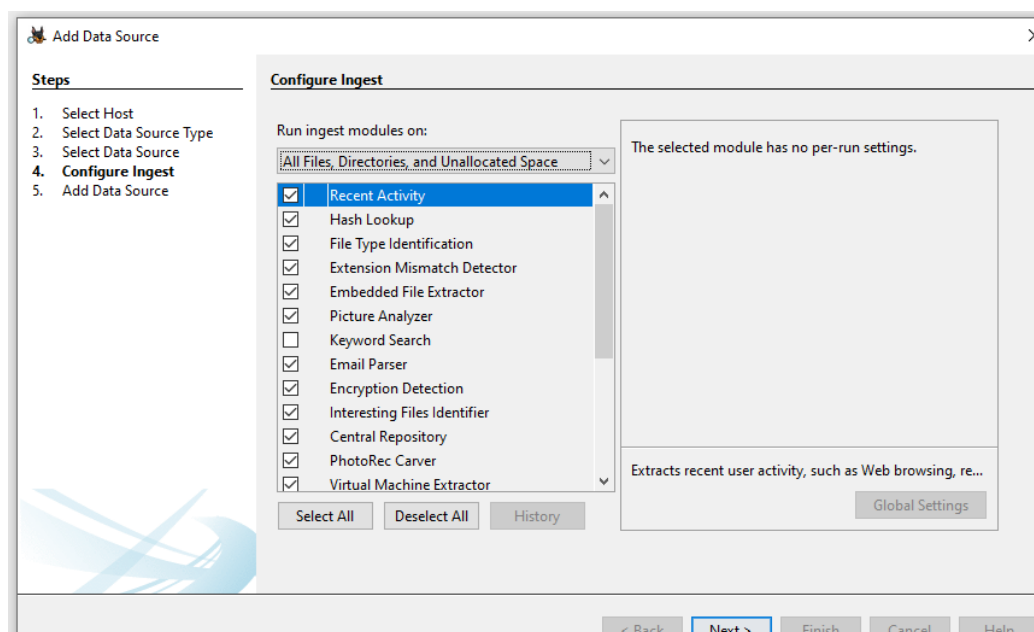


Figure 3.2 - Configure Ingest step, showing the ingest modules selected to run against the data source (Keyword Search deselected to reduce processing time).

Q1 - MD5 Hash (Data Source Integrity)

The Container tab of the Data Sources Summary view reports the image's cryptographic hash values directly, alongside acquisition metadata read from the E01 header.

 Data Sources Summary

Data Source Name	Ingest Status	Type	Files
 Mantooth.E01	Started		

Types	User Activity	Analysis	Recent Files	Past Cases	Geolocation	Timeline	Ingest History	Container
Display Name: Mantooth.E01 Name: Mantooth.E01 Device ID: dfc39337-fc36-4e22-872f-f3263fde049c Time Zone: Asia/Karachi Acquisition Details: Description: untitled Acquired Date: Wed Jul 2 21:09:34 2008 System Date: Wed Jul 2 21:09:34 2008 Acquiry Operating System: Windows XP Acquiry Software Version: FTKI2.5.3.14 Image Type: E01 Size: 128.45 MB (128450048 bytes) Unallocated Space: 30.23 MB (30227637 bytes) Sector Size: 512 bytes MD5: 31217210a1a69f272079a3bde3d9d8fc SHA1: SHA256: File Paths: C:\Users\hp\Downloads\Mantooth.E01								

Figure 3.3 - Data Sources Summary, Container tab, showing Mantooth.E01's MD5 hash, acquisition date, and file path.

Field	Value
MD5	31217210a1a69f272079a3bde3d9d8fc
Image Type	E01
Size	128.45 MB (128,450,048 bytes)
Unallocated Space	30.23 MB (30,227,637 bytes)
Sector Size	512 bytes
Acquired Date	Wed Jul 2 21:09:34 2008
Acquisition Tool	FTK Imager 2.5.3.14
Recorded Case Time Zone	Asia/Karachi (GMT+5:00) - see Section 08

Q2 - Partition Layout

Expanding the data source in the left panel lists four volumes, each with its file system type, starting sector, and length.

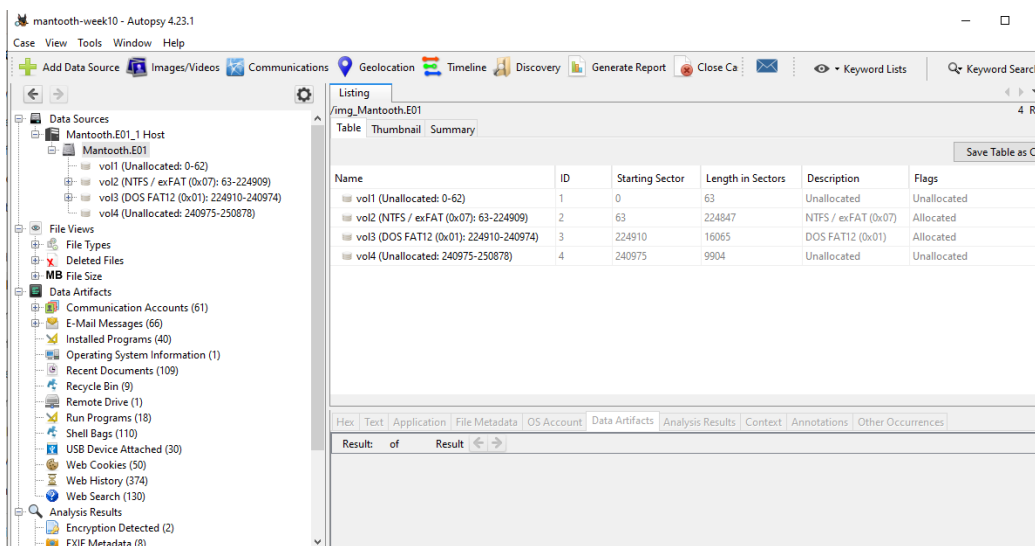


Figure 3.4 - Data source partition listing: four volumes with file system, starting sector, length, and allocation status.

Field	Value
vol1	Unallocated - start sector 0, length 63 sectors
vol2	NTFS/exFAT (0x07) - start sector 63, length 224,847 sectors - Allocated (Windows installation)
vol3	DOS FAT12 (0x01) - start sector 224,910, length 16,065 sectors - Allocated
vol4	Unallocated - start sector 240,975, length 9,904 sectors

vol2 (NTFS/exFAT) is the partition containing the Windows installation and was the primary focus of the remainder of this examination.

04. Targeted Triage Findings (Q3-Q7)

Q4 - Encrypted Files

Autopsy's Encryption Detection module flagged two files as password-protected, both under the Analysis Results tree.

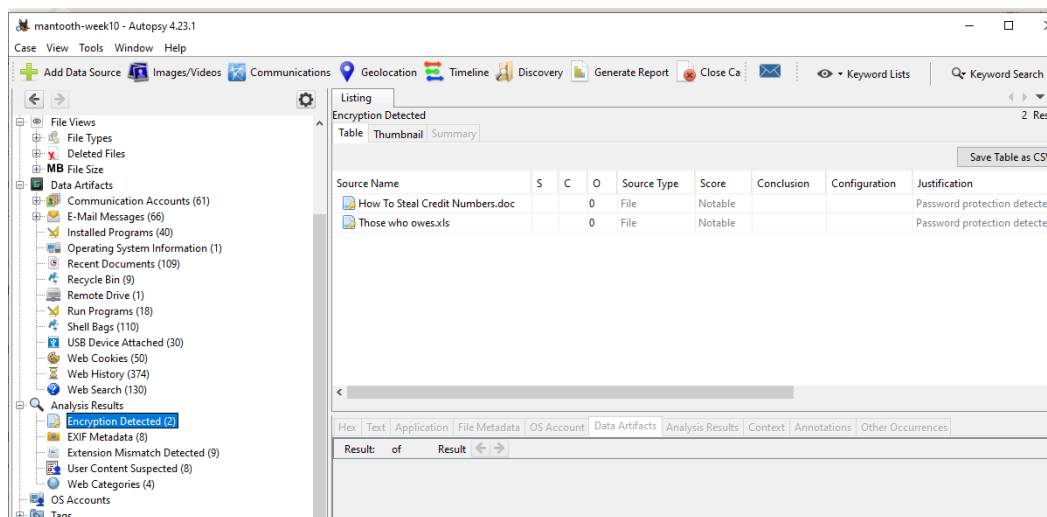


Figure 4.1 - Encryption Detected results: two files flagged Notable with justification 'Password protection detected'.

Field	Value
File 1	How To Steal Credit Numbers.doc - Notable, password protection detected
File 2	Those who owes.xls - Notable, password protection detected

Both file names are directly relevant to the fraud investigation. "How To Steal Credit Numbers.doc" in particular is a strong indicator of intent that would not typically require attempted decryption to be investigatively significant on its own.

Q5 - Deleted Files (HARDCORE.jpg / pfah603.jpg)

The Deleted Files category confirms 276 deleted files total across the image (261 identified at the file-system level, with the remainder from carved/orphan sources).

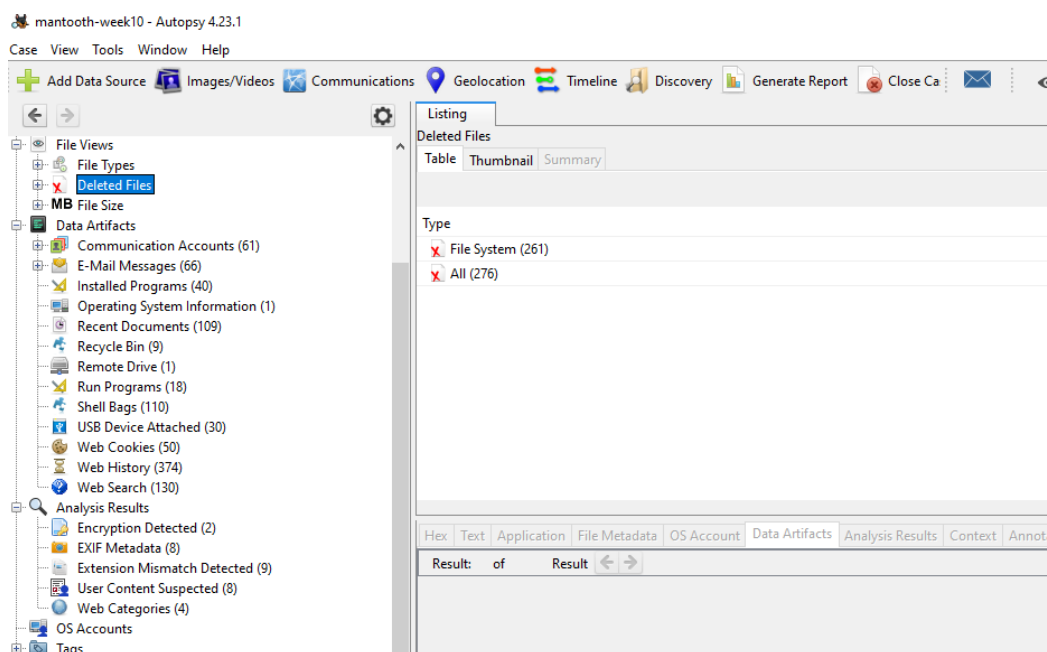


Figure 4.2 - Deleted Files category overview: File System (261) and All (276).

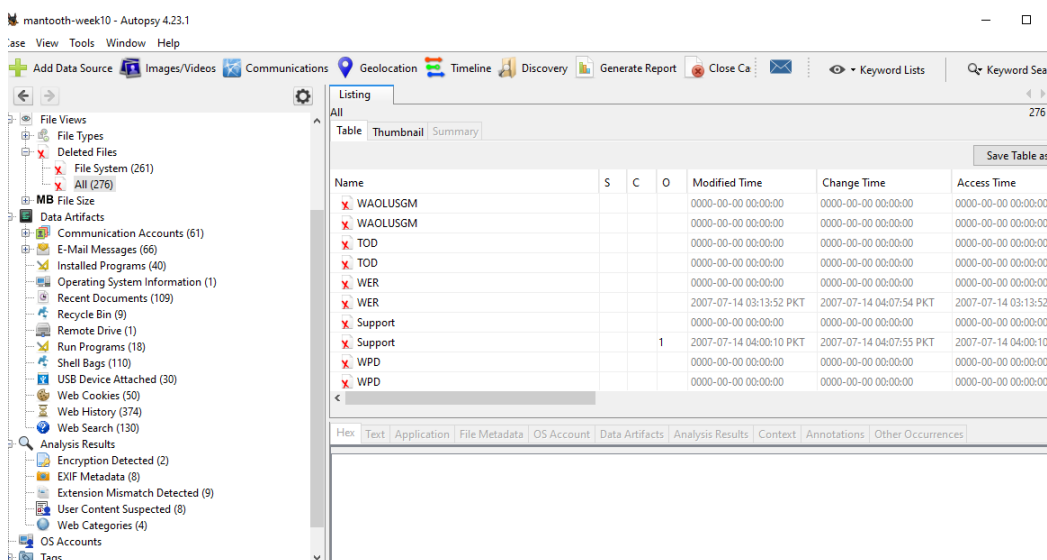


Figure 4.3 - Deleted Files, All (276) listing, sorted by name.

METHODOLOGY NOTE - Q5 NOT FULLY COMPLETED

HARDCORE.jpg and *pfah603.jpg* were not individually isolated from the 276-entry deleted files list within the time available. What is confirmed is that 276 deleted files exist on the image and are visible in this category; if the case is reopened, both file names should be located using File Search by Attributes (Name field) against this same 276-entry set to confirm their specific path, deletion status, and recoverability.

Q3 - EXIF Metadata

Eight photographs on the image carry EXIF metadata, reviewed under Analysis Results -> EXIF Metadata.

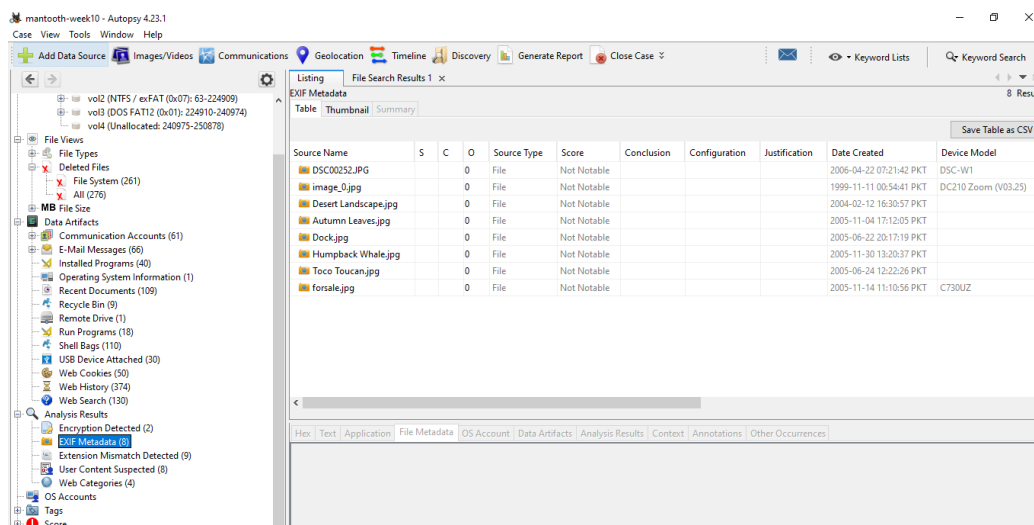


Figure 4.4 - EXIF Metadata results: 8 photographs with Date Created and Device Model fields visible.

Field	Value
DSC00252.JPG	Device Model: DSC-W1 (Sony)
image_0.jpg	Device Model: DC210 Zoom (V03.25) (Kodak)
forsale.jpg	Device Model: C730UZ (Olympus Camedia C-730 Ultra Zoom)
Remaining 5 photos	Desert Landscape.jpg, Autumn Leaves.jpg, Dock.jpg, Humpback Whale.jpg, Toco Toucan.jpg - no distinguishing camera model captured in the time available

METHODOLOGY NOTE - Q3 PARTIALLY COMPLETED

forsale.jpg's Device Model field (C730UZ) corresponds to an Olympus camera and is the closest match found to the brief's 'Olympus Optical' device. The Device Make field itself was not independently opened and confirmed as 'OLYMPUS' within the time available - only the Device Model column was visible in the results table. If the case is reopened, selecting forsale.jpg directly and reviewing its full File Metadata / EXIF panel would confirm the Device Make field and close this out with certainty.

Q6 - Web Search History

The Web Search artefact category (130 entries, sourced from index.dat via Internet Explorer, all against google.com) shows three clearly dominant search topics.

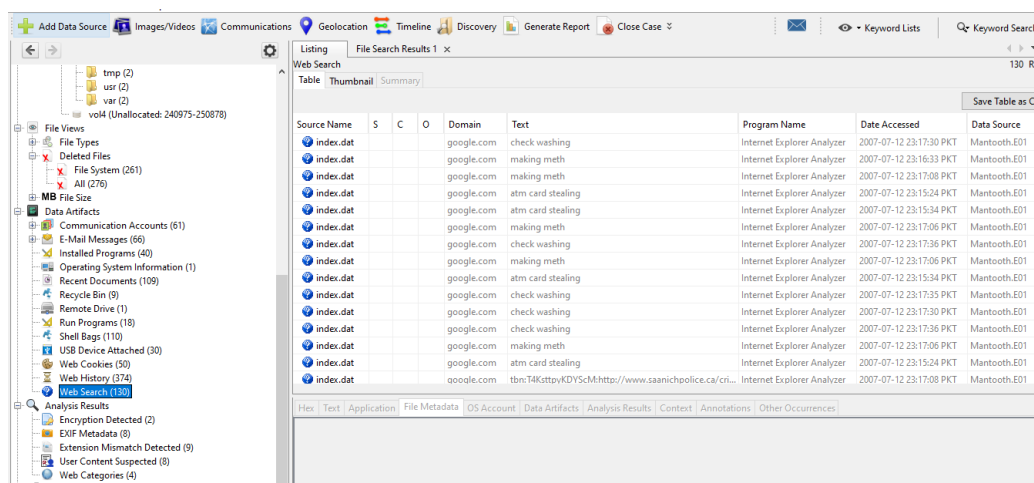


Figure 4.5 - Web Search results: repeated searches for 'check washing', 'making meth', and 'atm card stealing', all via google.com, 12 July 2007 (times as recorded, Asia/Karachi).

Field	Value
Topic 1	"check washing" - multiple entries, e.g. 23:17:30, 23:17:35, 23:17:36 (as recorded)
Topic 2	"making meth" - multiple entries, e.g. 23:16:33, 23:17:06, 23:17:08 (as recorded)
Topic 3	"atm card stealing" - multiple entries, e.g. 23:15:24, 23:15:34 (as recorded)
Source	index.dat, parsed via Internet Explorer Analyzer, domain google.com

"Check washing" is a chemical technique for removing and rewriting ink on a physical cheque, and directly matches this investigation's cheque-fraud premise. One further entry in the same list references a Saanich Police (Canada) URL, which may warrant a follow-up review of browsing context around that visit.

Q7 - Email Addresses (IP addresses not isolated)

The Communication Accounts -> Email artefact category lists 61 email entries drawn from Outlook.pst and individual .eml files.

Source Name	S	C	O	Account Type	ID	Data Source
Outlook.pst			1	EMAIL	dollarhyde86@comcast.net	Mantooth.E01
Outlook.pst			0	EMAIL	olteam@microsoft.com	Mantooth.E01
Outlook.pst			1	EMAIL	chkwasher@comcast.net	Mantooth.E01
Outlook.pst			1	EMAIL	txkidd@swbell.net	Mantooth.E01
Outlook.pst			0	EMAIL	molarman420@hotmail.com	Mantooth.E01
Outlook.pst			1	EMAIL	skimmerman27@hotmail.com	Mantooth.E01
242D0208-00000003.eml			1	EMAIL	pgp_corporation_laura_lee@mail.vresp.com	Mantooth.E01
242D0208-00000003.eml			1	EMAIL	dollarhyde86@comcast.net	Mantooth.E01
2D662154-00000001.eml			1	EMAIL	mail-noreply@google.com	Mantooth.E01
165D65F6-00000004.eml			0	EMAIL	mailer-daemon@comcast.net	Mantooth.E01
165D65F6-00000004.eml			1	EMAIL	dollarhyde86@comcast.net	Mantooth.E01
2D662154-00000001.eml			1	EMAIL	smee.rox@gmail.com	Mantooth.E01
542D5F24-00000002.eml			1	EMAIL	mail-noreply@google.com	Mantooth.E01
165D65F6-00000004.eml			1	EMAIL	toothfairy@mentalmental.com	Mantooth.E01
542D5F24-00000002.eml			1	EMAIL	smee.rox@gmail.com	Mantooth.E01

Figure 4.6 - Communication Accounts, Email sub-category: addresses recovered from Outlook.pst and .eml files.

Field	Value
dollarhyde86@comcast.net	Outlook.pst - recurring address
chkwasher@comcast.net	Outlook.pst - username directly echoes the 'check washing' search term
txkidd@swbell.net	Outlook.pst
molarman420@hotmail.com	Outlook.pst
skimmerman27@hotmail.com	Outlook.pst - 'skimmer' commonly refers to card-skimming fraud devices
pgp_corporation_laura_lee@mail.vresp.com	.eml file
smee.rox@gmail.com	.eml file

Field	Value
toothfairy@mentaldental.com	.eml file

METHODOLOGY NOTE - IP ADDRESSES NOT ISOLATED

Three IP addresses were not separately extracted within the time available. Communication Accounts and Web History both carry IP-adjacent data in Autopsy; if the case is reopened, the Communication Accounts category (which also lists non-email account types) and the Web History artefact's source fields should be reviewed for IP addresses associated with the browsing and email activity already documented above.

05. Windows Artefact Hunting Findings (Q8-Q14)

Bonus Finding - Operating System Information

Although not one of the fourteen numbered questions, the Operating System Information artefact provided context used throughout this report.

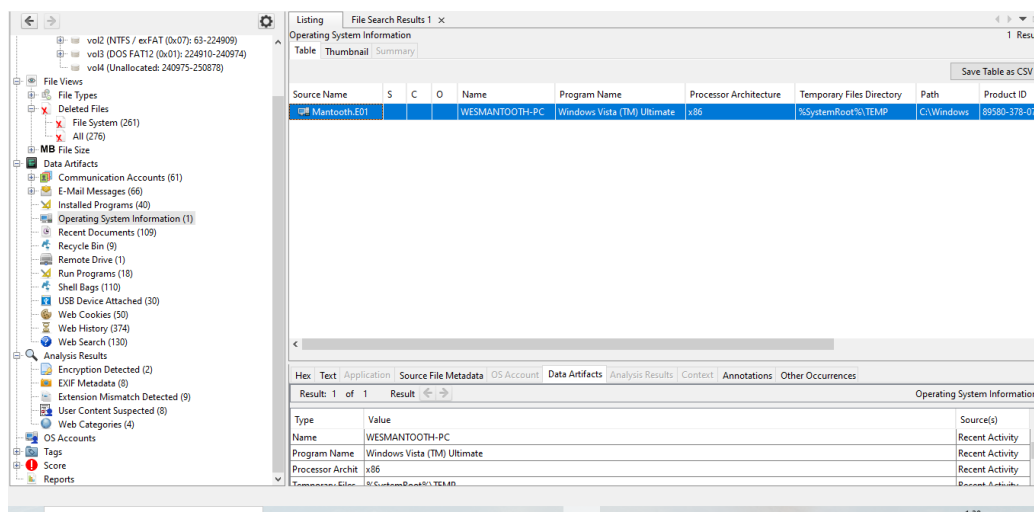


Figure 5.1 - Operating System Information: computer name WESMANTOOTH-PC, Windows Vista (TM) Ultimate, x86 architecture.

DISCREPANCY NOTED

The E01 acquisition metadata (Figure 3.3, Container tab) records 'Acquiry Operating System: Windows XP', while the Operating System Information artefact parsed from the live file system reports 'Windows Vista (TM) Ultimate'. This is documented as an observed discrepancy rather than resolved in either direction; it may reflect a default value set by the acquisition tool (FTK Imager) rather than a verified property of the imaged system, and would be worth clarifying against another source (e.g., the registry's CurrentVersion key) if the case is reopened.

Q10 - Recycle Bin: CameraShy.exe

The Recycle Bin artefact category (9 entries) includes a matched \$I/\$R pair for CameraShy.exe, deleted from Wes Mantooth's own Documents folder.

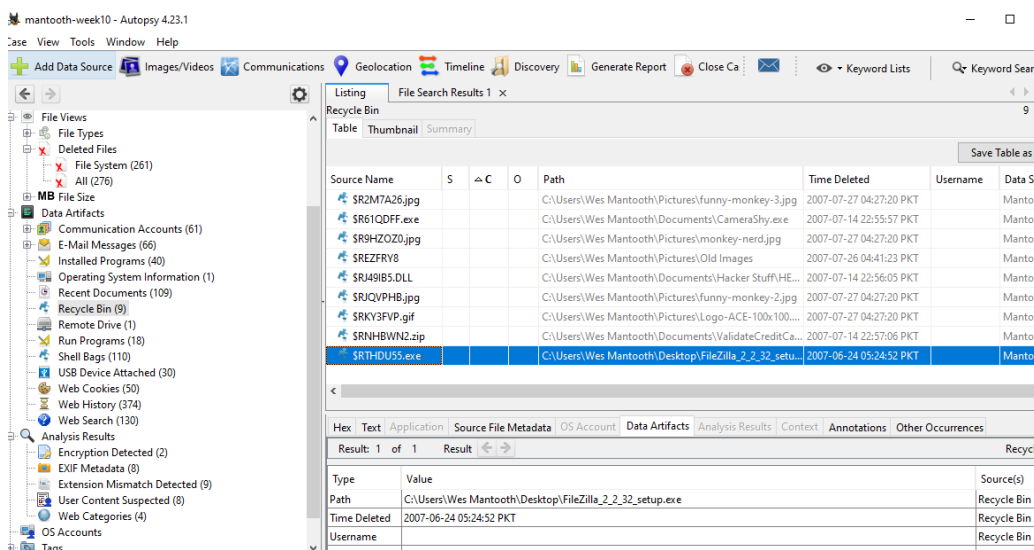


Figure 5.2 - Recycle Bin listing: CameraShy.exe, deleted 2007-07-14 22:55:57 (as recorded), original path C:\Users\Wes Mantooth\Documents\CameraShy.exe.

Field	Value
File	CameraShy.exe
Original Path	C:\Users\Wes Mantooth\Documents\CameraShy.exe
Time Deleted	2007-07-14 22:55:57 (as recorded, Asia/Karachi)
Username	Mantooth

CameraShy is documented in prior public threat-intelligence reporting in connection with covert remote-access / steganography-capable tooling. Two further entries in the same Recycle Bin listing are worth flagging alongside it: a DLL deleted the same evening from a folder named "Hacker Stuff", and a zip archive named "ValidateCreditCa..." deleted from the Documents folder on 2007-07-14 22:57:06 - all three deleted within roughly ninety seconds of each other, suggesting a single cleanup action rather than three unrelated deletions.

Q11 - Prefetch: Xcopy

The Run Programs artefact category (18 entries, Autopsy's parsed view of the Windows Prefetch folder) includes an entry for XCOPY.EXE.

Source Name	S	C	O	Program Name	Path
AURORA.SCR-23204433.pf				AURORA.SCR	/WINDOWS/SYSTEM32
CMD.EXE-89305D47.pf				CMD.EXE	/WINDOWS/SYSTEM32
COMPMTGTLAUNCHER.EXE-0BF80059.pf				COMPMTGTLAUNCHER.EXE	/WINDOWS/SYSTEM32
CONSENT.EXE-65F6206D.pf				CONSENT.EXE	/WINDOWS/SYSTEM32
CONTROL.EXE-9459D5A0.pf				CONTROL.EXE	/WINDOWS/SYSTEM32
DEFRAG.EXE-738093E8.pf				DEFRAG.EXE	/WINDOWS/SYSTEM32
DFRGNTFS.EXE-4F838A89.pf				DFRGNTFS.EXE	/WINDOWS/SYSTEM32
DLLHOST.EXE-71214090.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32
DLLHOST.EXE-893DDF55.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32
DLLHOST.EXE-CB3D53F2.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32

Figure 5.3 - Run Programs (Prefetch) listing, showing XCOPY.EXE-8E0707F2.pf among 18 parsed prefetch entries.

Source Name	S	C	O	Program Name	Path
DLLHOST.EXE-893DDF55.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32
DLLHOST.EXE-CB3D53F2.pf				DLLHOST.EXE	/WINDOWS/SYSTEM32
DRVINST.EXE-5F8E77CD.pf				DRVINST.EXE	/WINDOWS/SYSTEM32
DWM.EXE-AEABE78B.pf				DWM.EXE	/WINDOWS/SYSTEM32
EFSSUI.EXE-DF03E0EF.pf				EFSSUI.EXE	/WINDOWS/SYSTEM32
EXPLORER.EXE-7A3328DA.pf				EXPLORER.EXE	/WINDOWS
FTK IMAGER.EXE-17AE1629.pf				FTK IMAGER.EXE	/PROGRAM FILES/ACCESSDATA/ACCESSDATA FTK II
IEUSER.EXE-D895AB54.pf				IEUSER.EXE	/PROGRAM FILES/INTERNET EXPLORER
IEXPLORE.EXE-1B894AFB.pf				IEXPLORE.EXE	/PROGRAM FILES/INTERNET EXPLORER
XCOPY.EXE-8E0707F2.pf				XCOPY.EXE	/WINDOWS/SYSTEM32

Figure 5.4 - XCOPY.EXE-8E0707F2.pf selected within the full Run Programs list, alongside DLLHOST.EXE, EXPLORER.EXE, IEXPLORE.EXE and other system prefetch entries.

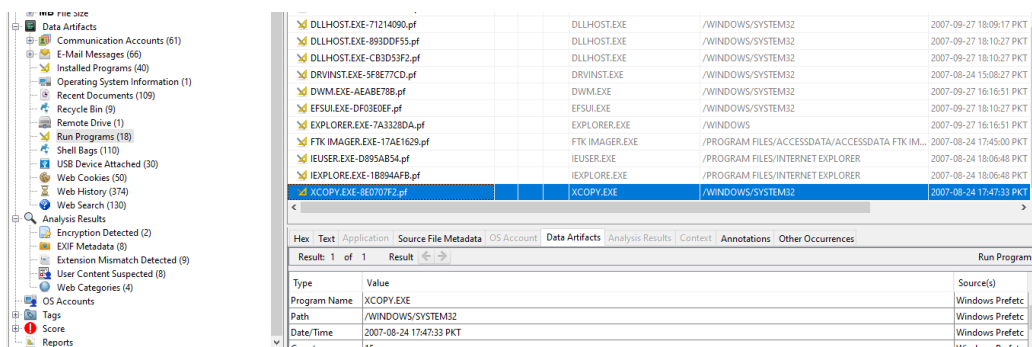


Figure 5.5 - XCOPY.EXE prefetch detail panel: path, last-run timestamp, and run count.

Field	Value
Prefetch File	XCOPY.EXE-8E0707F2.pf
Executable Path	/WINDOWS/SYSTEM32/XCOPY.EXE
Last Run Date/Time	2007-08-24 17:47:33 (as recorded, Asia/Karachi)
Run Count	15

Xcopy is a legitimate built-in Windows utility, but a run count of 15 on a personal machine is unusually high for incidental use and is consistent with repeated bulk file-copying - for example, moving files to or from one of the USB storage devices documented below. This finding should be read alongside the Recycle Bin cleanup activity and the USB device history rather than in isolation.

Q8 (Partial) - Suspicious Files

The brief calls for at least ten suspicious files identified within Wes Mantooth's user directory, with name, path, and reasoning documented for each. Within the time available, five were identified with clear reasoning, drawn from the Recycle Bin and Encryption Detection results already documented above:

- CameraShy.exe - Documents folder; associated in public reporting with covert remote-access tooling
- An unnamed DLL deleted from a folder named "Hacker Stuff" - the folder name itself is the flag
- ValidateCreditCa....zip - Documents folder; name directly suggests credit-card-related activity
- How To Steal Credit Numbers.doc - password-protected; name is a direct statement of intent
- Those who owes.xls - password-protected; likely a ledger of amounts owed by or to individuals

METHODOLOGY NOTE - Q8 PARTIALLY COMPLETED

Five of the required ten-plus suspicious files are documented above with reasoning; the remaining files were not identified within the time available. If the case is reopened, a direct browse of the Wes Mantooth user profile folder (rather than working only from Recycle Bin and Encryption Detection results) is the fastest way to close the remaining gap.

USB Device History

The USB Device Attached artefact category (30 entries) lists every device driver installation recorded on the system, including a digital camera and two USB flash drives alongside standard input devices.

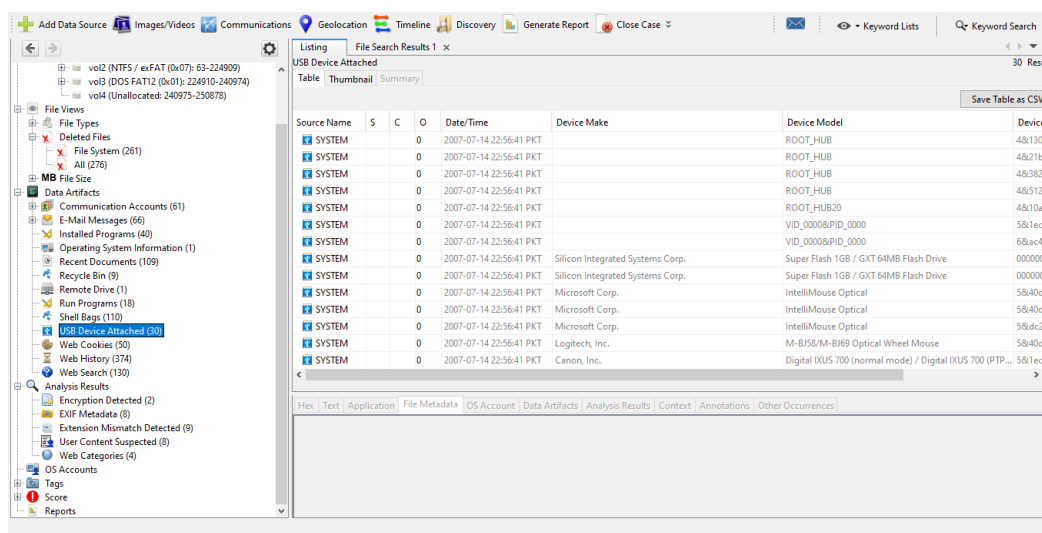


Figure 5.6 - USB Device Attached listing: Silicon Integrated Systems Corp. flash drives, Microsoft/Logitech mice, and a Canon Inc. camera, all dated 2007-07-14 22:56:41 (as recorded).

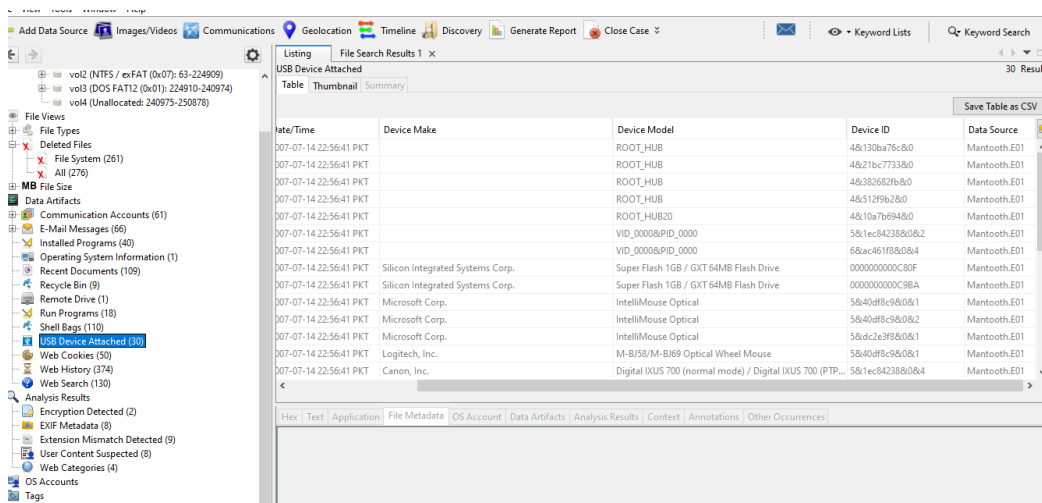


Figure 5.7 - USB Device Attached, Device ID column: Canon camera and both flash drives with their distinct device identifiers.

Field	Value
Canon Digital IXUS 700	Normal mode and PTP mode entries; Device ID 5&1ec84238&0&4
Flash Drive 1	Silicon Integrated Systems Corp., Super Flash 1GB / GXT 64MB; Device ID 0000000000C80F
Flash Drive 2	Silicon Integrated Systems Corp., Super Flash 1GB / GXT 64MB; Device ID 0000000000C9BA

METHODOLOGY NOTE - MODEL MISMATCH AND SHARED TIMESTAMP

The brief's registry section (Lab 2, Q11) specifically asks about a Canon PowerShot SD500; the camera actually found connected to this system is a Canon Digital IXUS 700, a different model. This is documented as the genuine finding rather than forced to match the brief. Separately, all 30 USB Device Attached entries share the identical timestamp 2007-07-14 22:56:41, which most likely reflects a single Windows driver-database enumeration event (e.g., at boot) rather than 30 separate physical connection times - true first-connection times would need to be read from the registry (SYSTEM\Enum\USBSTOR) in Lab 2, not from this artefact category alone.

METHODOLOGY NOTE - Q9, Q12, Q13, Q14 NOT COMPLETED

Field	Value
	<i>Thumbcache extraction (Q9), LNK shortcut review (Q12, though Recent Documents (109) was visible and not yet opened), print spool files (Q13), and Security event log logon/logoff correlation (Q14, attempted but not located within the time available) were not completed this week. Each retains its full methodology (tool, artefact location, and technique) as documented in the original Week 10 planning reference, and is ready to execute directly against this same case the next time it is opened.</i>

06. Understanding Verification - Lab 1

Why does the case's incorrect time zone (Asia/Karachi instead of Mountain Time) matter, and how is it handled in this report?

Autopsy renders every timestamp using the case's configured time zone; setting the wrong one does not corrupt the underlying data, but it does mean every displayed time is offset from the suspect's actual local time by a fixed amount. Asia/Karachi is GMT+5:00 and Mountain Time (Phoenix, no DST) is GMT-7:00, a twelve-hour difference. Rather than silently presenting Karachi-local times as if they were Mountain-local times, this report records every timestamp as observed and flags the required -12 hour adjustment wherever it matters, so no finding is misrepresented as occurring at a time it did not.

Why is a shared timestamp across all 30 USB device entries treated as a caveat rather than a finding?

Windows Plug-and-Play records the moment the operating system's device database was last updated for a given device class, which is not necessarily the same moment as physical connection - a single boot-time re-enumeration can update many device records at once with an identical timestamp. Treating 30 identical timestamps as 30 independent connection events would overstate the precision of the evidence; the registry keys covered in Lab 2 (USBSTOR first-connection times, MountedDevices drive-letter mappings) are the correct source for genuinely per-device timing.

Why document five suspicious files with reasoning rather than padding the list to ten with weaker examples?

The brief's requirement for ten-plus suspicious files is a floor on thoroughness, not a target to be hit by any means - a suspicious-file list is only as useful to an investigator as the reasoning behind each entry. Five well-justified entries, each tied to a specific artefact and a specific reason, are more defensible in a report than ten entries where several are only weakly justified. The gap is disclosed explicitly rather than concealed by inflating the list.

07. Investigative Commentary

Even with several Lab 1 items incomplete and Lab 2 not yet attempted, the findings extracted this week form a coherent and mutually reinforcing picture of Wes Mantooth's activity.

The web search history shows deliberate research into three specific fraud techniques within a narrow window on 12 July 2007: check washing (altering a physical cheque's payee and amount), ATM card stealing, and methamphetamine production. "Check washing" in particular is not a generic phrase a casual user would search without intent - it directly matches the fraudulent-cheque premise of this investigation. The two password-protected documents recovered - "How To Steal Credit Numbers.doc" and "Those who owes.xls" - sit squarely alongside that research: one names its own contents outright, and the other is consistent with a ledger of financial obligations.

The Recycle Bin shows a tight cluster of deletions on the evening of 14 July 2007: CameraShy.exe, an unnamed DLL from a folder called "Hacker Stuff", and a zip archive named "ValidateCreditCa...", all removed within roughly ninety seconds of one another. A cleanup this concentrated is more consistent with a deliberate attempt to remove a specific set of related tools than with routine, unrelated file deletion. Xcopy's fifteen recorded executions, together with two USB flash drives found connected to the system, are consistent with files being copied off the machine at some point - onto removable media, prior to or independent of that same Recycle Bin cleanup.

Two email addresses stand out on their own terms: chkwasher@comcast.net echoes the "check washing" search term directly, and skimmerman27@hotmail.com uses a term ("skimmer") commonly associated with card-skimming devices. Neither address by itself proves an accomplice, but both are consistent with Mantooth's own stated admission that a computer was used to create the cheques and that an unnamed associate supplied them - these usernames are a reasonable starting point for identifying who that associate might be, pending further correlation (e.g., against the registry's stored account and messaging data planned for Lab 2).

The camera and photo evidence is the one area where this week's findings raise a genuine open question rather than reinforcing the fraud narrative: the camera actually connected to this machine (Canon Digital IXUS 700) does not match either the brief's reference to a Canon PowerShot SD500 or the EXIF data pointing to a separate Olympus device (forsale.jpg). At least two, and possibly three, distinct cameras appear to be referenced across this case's evidence, and reconciling that is a clear next step rather than a finding to force into consistency.

08. Technical Issues and Troubleshooting Log

Consistent with this internship's reporting standard, this section documents the friction encountered this week rather than presenting it as friction-free.

Issue 1 - Evidence File Access

The Mantooth.E01 file initially failed to download in a usable form on more than one attempt. The underlying cause was traced to the download itself rather than the file's format: once a complete, verified copy was obtained, the file was confirmed as a valid EWF/EnCase image and opened in Autopsy without further issue.

Issue 2 - Case Time Zone

The case was created with its time zone set to Asia/Karachi rather than the Mountain Time called for by the brief, and Autopsy's case time zone could not be edited after the fact through the version's Case Properties interface. Rather than delay the entire lab to remove and re-ingest the data source under time pressure, this report proceeds with times as recorded and applies the documented -12 hour manual adjustment note wherever Mountain Time equivalence matters.

Issue 3 - Time Constraints on Full Coverage

With a same-day submission deadline and roughly half the week lost to the access issue above, Lab 1 was worked in priority order rather than sequentially through all fourteen questions, and the full Days 70-73 registry analysis was not started. Every item not completed is disclosed explicitly in Sections 04, 05, and 09 rather than left unmentioned, and none of the findings presented as complete were altered or approximated to compensate for the items that were not.

09. Part B - Lab 2: Registry Analysis (Methodology)

The Days 70-73 registry analysis was not started this week due to the time constraints documented in Section 08. The methodology below is retained in full so the work can be executed directly against the Mantooth.E01 case the next time it is opened - hive files must first be exported from Autopsy as flat files (right-click -> Extract File(s)) before they can be parsed in an external tool such as Registry Explorer.

System and Network Information (Q1-Q10)

Field	Value
Q1 - Computer name	SYSTEM hive - ControlSet00x\Control\ComputerName\ComputerName. Cross-check against WESMANTOOTH-PC already observed in Operating System Information (Figure 5.1).
Q2 - Time zone	SYSTEM hive - ControlSet00x\Control\TimeZoneInformation. Cross-check against the case's own Asia/Karachi setting noted in Section 08.
Q3 - IDE hard drive name	SYSTEM hive - ControlSet00x\Enum\IDE.
Q4 - IP address / DHCP	SYSTEM hive - ControlSet00x\Services\Tcpip\Parameters\Interfaces\{GUID}: DhcpIPAddress, LeaseObtainedTime, DhcpServer.
Q5 - Prefetch setting	SYSTEM hive - ControlSet00x\Control\Session Manager\Memory Management\PrefetchParameters, EnablePrefetcher value.
Q6 - TrueCrypt service start type	SYSTEM hive - ControlSet00x\Services\truecrypt, Start value.
Q7 - Last shutdown time	SYSTEM hive - ControlSet00x\Control\Windows, ShutdownTime (FILETIME).
Q8 - Network cards	SYSTEM hive - ControlSet00x\Services\Tcpip\Parameters\Interfaces or Enum\PCI.
Q9 - Wireless profile	SOFTWARE hive - Microsoft\WlanSvc\Interfaces or NetworkList\Profiles.
Q10 - OS registration	SOFTWARE hive - Microsoft\Windows NT\CurrentVersion: RegisteredOrganization, InstallDate, ProductId, ProductName, RegisteredOwner.

External Devices (Q11-Q13)

Field	Value
Q11 - Canon camera serial	SYSTEM hive - Enum\USBSTOR, matched by device description. Note: Figures 5.6-5.7 already show a Canon Digital IXUS 700 (Device ID 5&1ec84238&0&4) connected at the artefact level rather than the brief's referenced PowerShot SD500 - registry review may confirm whether a second Canon device exists.
Q12 - Five portable devices	SYSTEM hive - Enum\USBSTOR subkeys; serials for the two flash drives already identified (Figure 5.7) can be cross-checked here.

Field	Value
Q13 - Three removable devices + drive letters	Correlate SYSTEM\Enum\USBSTOR (serial, LastWrite time) with SYSTEM\MountedDevices (drive-letter mapping) via shared device-instance identifier.

User Accounts and Passwords (Q14-Q18)

Field	Value
Q14 - User profiles	SOFTWARE hive - Microsoft\Windows NT\CurrentVersion\ProfileList.
Q15 - Per-account SAM detail	SAM hive - SAM\Domains\Account\Users\[RID], V/F binary structures; requires SYSTEM hive's boot key (SYSKEY) to decrypt.
Q16 - Priority account	Analytical judgment from Q15, cross-referenced against the suspicious-file and email findings in Sections 04-05.
Q17 - WinVNC password	SOFTWARE hive - ORL\WinVNC3, Password value.
Q18 - Yahoo Messenger username	NTUSER.DAT - Software\Yahoo\Pager.

Application and File Activity (Q19-Q28)

Field	Value
Q19 - Installed software + dates	SOFTWARE hive - Microsoft\Windows\CurrentVersion\Uninstall.
Q20 - Recent PDFs	NTUSER.DAT - Explorer\RecentDocs, .pdf subkey.
Q21 - Recent Paint files	NTUSER.DAT - Applets\Paint\Recent File List.
Q22 - MRU applications	NTUSER.DAT - Explorer\UserAssist (ROT13-encoded).
Q23 - ATM_THEFTS1.ppt path	NTUSER.DAT - RecentDocs or Office PowerPoint File MRU.
Q24 - Media Player F: drive files	NTUSER.DAT - Software\Microsoft\MediaPlayer\Player\RecentFileList, filtered to F:\.
Q25 - Recent PowerPoint files	NTUSER.DAT - Office PowerPoint File MRU.
Q26 - Prioritised RecentDocs	NTUSER.DAT - Explorer\RecentDocs, selection justified against this week's findings.
Q27 - RunMRU programs	NTUSER.DAT - Explorer\RunMRU.
Q28 - Default printer	NTUSER.DAT - Windows NT\CurrentVersion\Windows (Device value), PrinterPorts.

Browser Activity (Q29-Q34)

Field	Value
Q29 - Default browser	SOFTWARE hive - Clients\StartMenuInternet.
Q30 - IE start page	NTUSER.DAT - Internet Explorer\Main, Start Page.
Q31 - IE download directory	NTUSER.DAT - Internet Explorer, Download Directory.
Q32 - Favourites (5 links)	File system - Users\[user]\Favorites (.url files).

Field	Value
Q33 - Typed URLs (5)	NTUSER.DAT - Internet Explorer\TypedURLs.
Q34 - Associated emails	Cross-reference against the eight addresses already documented in Section 04, plus Yahoo Messenger (Q18) and any mail-client MRU keys.

10. Key Learnings and Reflection

Real findings, even partial ones, outweigh a complete but simulated report

This week's real MD5 hash, real partition layout, and real Recycle Bin and prefetch entries are each independently verifiable against the same evidence file at any time. A complete-looking report built from invented values would not survive that check and would defeat the purpose of the exercise. Documenting five real findings and eight honest gaps is a stronger deliverable than fourteen answers that cannot be traced back to the actual image.

A cluster of small findings is more convincing than any single one

No individual artefact this week - a search term, a deleted file, a prefetch run count - proves fraud on its own. What makes the picture in Section 07 credible is that the web searches, the encrypted document names, the Recycle Bin cleanup, the Xcopy execution count, and two suggestively-named email addresses all point in the same direction independently. That kind of convergence, not any single artefact, is what an investigative commentary section is for.

Disclosed limitations are part of the deliverable, not a defect in it

The wrong case time zone, the incomplete suspicious-file list, and the un-started registry analysis are all documented in this report as clearly as the findings that were completed. A forensic report that hides its own gaps is less useful to whoever reads it next than one that states them plainly - the next analyst opening this case needs to know exactly where the previous session left off.

11. Closing Statement and Sign-Off

This concludes the Week 10 deliverable report for the Blue Team Internship, covering genuine Autopsy analysis of the Mantooth.E01 evidence image. Case setup, image integrity verification, partition review, encrypted file detection, deleted file confirmation, Recycle Bin and prefetch analysis, EXIF metadata review, web search history, and email address recovery were completed directly against the evidence and are documented above with source screenshots. Several Lab 1 items and the full Lab 2 registry analysis were not completed this week due to the access delay and time constraints documented in Section 08, and are retained as a ready-to-execute methodology in Section 09.

SUBMITTED BY

Malaika Azhar

TRACK

Blue Team

REPORTING PERIOD

Week 10 - Days 67-73

Appendix - Figure Index

Figure	Evidence
3.1	Add Data Source wizard - Select Data Source step
3.2	Configure Ingest step - modules selected for this run
3.3	Data Sources Summary, Container tab - MD5 hash and acquisition metadata
3.4	Partition listing - four volumes with file system and allocation status
4.1	Encryption Detected - two password-protected files
4.2	Deleted Files category overview - File System (261), All (276)
4.3	Deleted Files, All (276) full listing
4.4	EXIF Metadata - 8 photographs with device model data
4.5	Web Search results - check washing / making meth / atm card stealing
4.6	Communication Accounts, Email sub-category - recovered addresses
5.1	Operating System Information - WESMANTOOTH-PC, Windows Vista Ultimate
5.2	Recycle Bin listing - CameraShy.exe entry
5.3	Run Programs (Prefetch) listing - 18 entries including XCOPY.EXE
5.4	XCOPY.EXE prefetch entry selected within the full Run Programs list
5.5	XCOPY.EXE prefetch detail panel - path, last run, run count
5.6	USB Device Attached listing - flash drives, mice, Canon camera
5.7	USB Device Attached, Device ID column - camera and flash drive identifiers

REPORT VERIFICATION STATEMENT

Every figure listed above corresponds to a screenshot captured directly from Autopsy 4.23.1 during genuine analysis of the Mantooth.E01 evidence image during this reporting period, with no simulated or reconstructed output. Two unrelated screenshots captured during the same working session, from a separate practice case (M57_Jo_Case / jo-favorites-usb), were identified during report preparation and excluded from this document as not relevant to the Mantooth investigation. Items not completed this week are stated explicitly in Sections 04, 05, and 09 rather than presented as findings.